# MUSHABBAR HUSNAIN NOOR

## MECHANICAL ENGINEER


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## EDUCATION

**Bachelor of Science in Mechanical Engineering,** Air University, Islamabad September 2017-January 2022

## SKILLS & EXPERTISE

**Expertise:** Aerothermodynamics, Aeroelasticity, Optimization, Avionics Thermal Management, Turbomachinery, Design of Experiment (DoE)
**Software:** SOLIDWORKS, ANSYS (fluent, ICEM, Static Structural, thermal), MATLAB, Python(basic), COMSOL Multiphysics, Pointwise
**Prototyping:** Laser Cutting, Wire Cutting, CNC Lathe, CNC Mill, 3D Printing

## PROFESSIONAL EXPERIENCE

**Aerodynamics Engineer** – BAYKAR Technologies, Pakistan & Türkiye February 2024-Present

- Performed CFD analysis for unmanned aerial vehicles (UAVs) and propeller engines, optimizing aerodynamic efficiency and performance.
- Analyzed aerodynamic performance metrics, including static and dynamic stability parameters, to ensure flight stability and control.
- Calculating aerodynamics parameters for the development of hardware in loop system (HILS)
- Conducted fluid-structure interaction (FSI) based aeroelastic analysis to determine flutter boundaries and establish structural limits.
- Led the evaluation of design modifications, including G-load assessments on aircraft structures to enhance durability and performance.
- Developed comprehensive VN-diagrams and defined the flight envelope to determine operational safety and performance constraints.
- Performed structural ground testing of aerostructures, validating computational models
- Conducted modal analysis to identify and improve natural frequencies, enhancing structural integrity and mitigating resonance risks.

**Aerothermal Research Lead Engineer** – Shocks & Stars Engineering Pvt Ltd., Islamabad August 2022-January 2024

- Performing the aerothermal design analysis of airborne electronics at different flight regimes for **AEROFLUX III** project
- Conducting analytical and computational analysis to analyze the heat transfer of LRUs mounted in conditioned and unconditioned compartments of high-speed combat aircraft.
- Performing Computational Hypersonic Aerothermodynamic and combustion analysis of Scramjet Engine
- Computational analysis of Plasma Arc Thrusters using COMSOL Multiphysics
- Developing novel algorithms and high-fidelity math-models to solve the thermos-fluids problems.
- Applying computational design of experiment-based response surface method for advanced analysis of complete aircraft flight envelope
- Computational and analytical optimization of thermal parameters for optimal heat transfer design points
- To qualify the thermal design of avionics LRUs under MIL-STD-5400 at different altitude for PAF's Airworthiness Certification Authority
- Funding grant of PKR 1.5 million by SPRC, Govt, of Pakistan (Grant# # CO2SDM5N-OOSAT-OTS01234-220605)
- Funding grant by NESCOM (Agreement No. 083/RAC-IX/AU/2022)

**Aerothermal Lead Engineer –** AEROFLUX Lab, Air University**,** Islamabad January 2022-July 2022

- Performed the analytical and computational thermal airborne avionics system LRU placed in the conditioned bay of high performance aircraft for **AEROFLUX-II** project.
- Qualified the thermal design of avionics LRUs under MIL-STD-5400 for PAF's Airworthiness Certification Authority
- Thermal analysis of existing design under normal and failure mode for airborne electronics operations
- Designing multiple passive and active heat transfer mode for airborne electronics system
- Funding grant of PKR 2.5 million by SPRC, Govt, of Pakistan (Grant# CO2SDM5N-OOSAT-OTS02638-21084)

**Undergraduate Research Assistant –** AEROFLUX Lab, Air University**,** Islamabad June 2021-January 2022

- Conducted thermal analysis of airborne avionics system LRU placed in the conditioned bay of commercial aircraft.
- Performed the analytical and computational thermal analysis of **AEROFLUX I** project.
- Qualified the thermal design of avionics LRUs under MIL-STD-5400 at different for PAF's Airworthiness Certification Authority
- Thermal Analysis of existing design under normal and failure mode for LRU operations
- Designing multiple passive and active heat transfer mode
- Funding grant of PKR 1.5 million by SPRC, Govt, of Pakistan (Grant# CO2SDM5N-OKAIL-OTS00057-220404)

## PUBLICATIONS

- M. H. Noor, A. Sarosh, U. Riaz, M. Adnan and O. A. A. Awan, "A Figure of Merit Based Computational Heat Transfer Method for Aerothermal Analysis of Airborne Electronics System LRU," 2022 19th International Bhurban Conference on Applied Sciences and Technology (IBCAST), Islamabad, Pakistan, 2022, pp. 70-75, doi: 10.1109/IBCAST54850.2022.9990189.
- Noor, M.H., Sarosh, A., Khan, A.A. (2024). A Minimal Input Dimensionless Empirical-Corrector (MIDEC) Method for Aerothermal Analysis of Airborne Electronics. Proceedings of the First International Conference on Aeronautical Sciences, Engineering and Technology. ICASET 2023. Springer, Singapore. https://doi.org/10.1007/978-981-99-7775-8_20

## RESEARCH AND INDUSTRIAL PROJECTS

**AEROFLUX III** Shocks & Stars Engineering, Islamabad

**Aero-thermal design analysis of avionics systems LRUs mounted in conditioned nose bay of aircraft** June 2022–Present

- Performing analytical and computational (Ansys Fluent) aerothermal analysis of avionics systems
- Conducting transient analysis to assess the temperature behavior with respect to time.
- Determining the temperature behavior at different Mach numbers, flight altitudes and bleed flow settings of F100 PW 220E engine
- Qualifying the thermal design under MIL-E-5400

**AEROFLUX II** Air University, Islamabad

**Aero-thermal design analysis for qualification of airborne electronics LRU mounted in unconditioned dorsal bay of high performance combat aircraft** August 2021 – July 2022

- Determined the analytical and computational (Ansys Fluent) aerothermal analysis of avionics system under steady state conditions.
- Performed analysis at 30% and 100% duty cycle for different Mach number and altitudes at ISA and user defined conditioned.
- Applied the response surface based design of experiment method for the analysis of complete flight regime.
- Developed a high-fidelity algorithm in MATLAB using DoE resulted data for the temperature calculation of whole regime.
- Validated the solution using the test results of environmental thermal chamber.
- Qualified the thermal design of avionics system under MIL-E-5400

**AEROFLUX I** Air University, Islamabad

**Aero-thermal design analysis for qualification of airborne electronics LRU mounted in conditioned aircraft bays** June 2020–July 2021

- Conducted research for cooling of avionics components placed inside the air-conditioned bay of airborne platform.
- Performed analytical and computational thermal analysis of existing design under normal and failure mode for LRU operations.
- Verified the thermal results with CFD analysis.
- Qualified the thermal design of electronics under MIL-E-5400

## AWARDS & CERTIFICATES

- Certified Aerodynamics Engineer for Loiter Munition UAVs By Baykar Technologies
- Honorable Judge at CAD Mania Competition of Air University Islamabad
- Pakistan Engineering Council (PEC) Certified Engineer
- Winner of the Air Cad Competition in Airtech'19
- PEEF Scholarship (2017-2021)

## LANGUAGES

- English (Fluent)
- Urdu (Native)